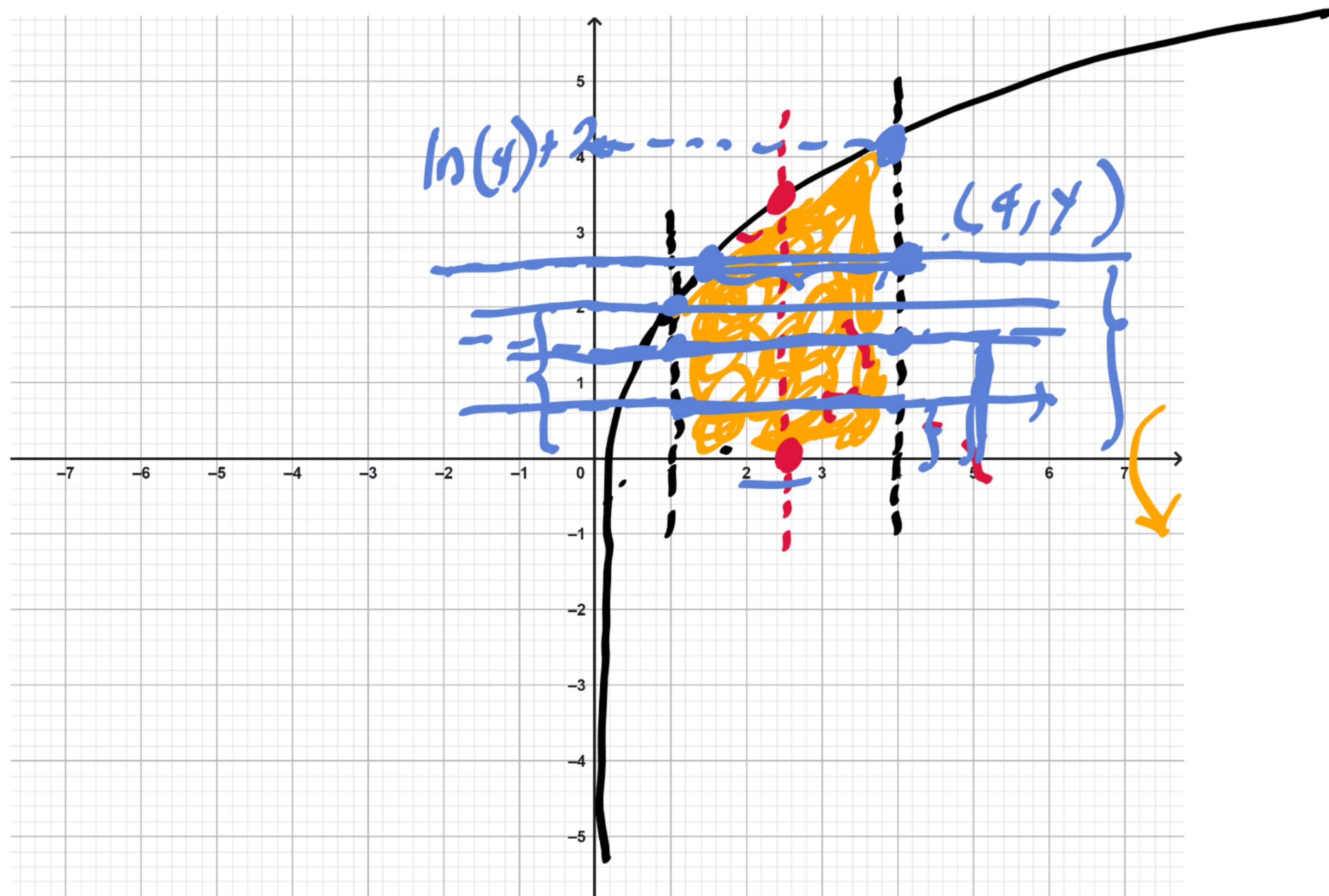


Region is closed by

$$\begin{cases} y = \ln(x) + 2 \\ x = 1 \\ x = 4 \\ x\text{-axis} \end{cases}$$



$$\text{Area} = \int_1^4 (\ln(x) + 2) dx$$

$$V_{\text{Disk}} = \int_1^4 \pi (\ln(x) + 2)^2 dx$$

$$V_S = \int_0^2 2\pi (y) \underbrace{(3)}_{4-1} dy + \int_2^{\ln(4)+2} 2\pi (y) (4 - e^{y-2}) dy$$

$$y = \ln(x) + 2$$

$$y - 2 = \ln(x)$$

$$x = e^{y-2}$$

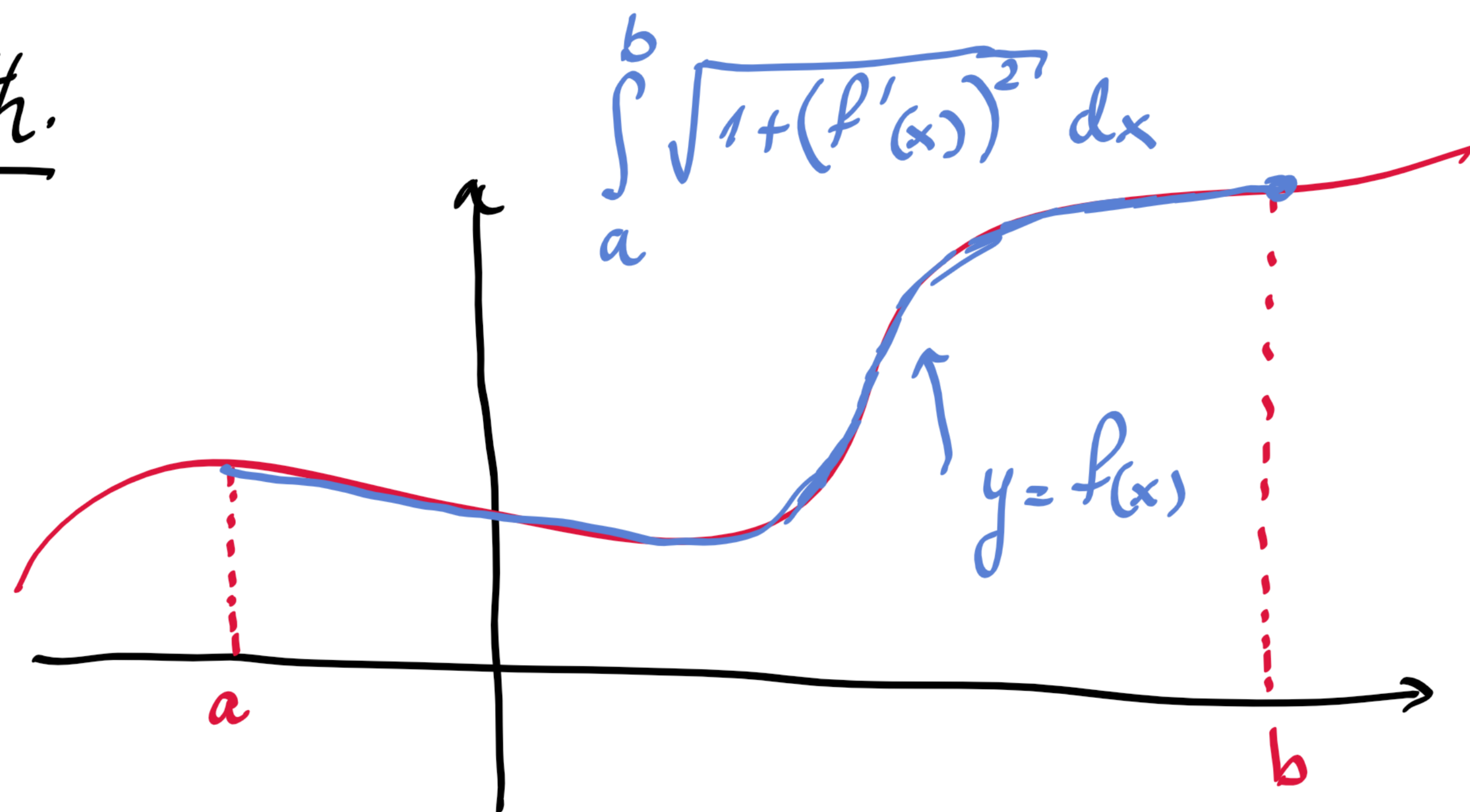
$$\int_1^4 \ln(x) dx$$

$$\int_1^4 \ln^2(x) dx$$

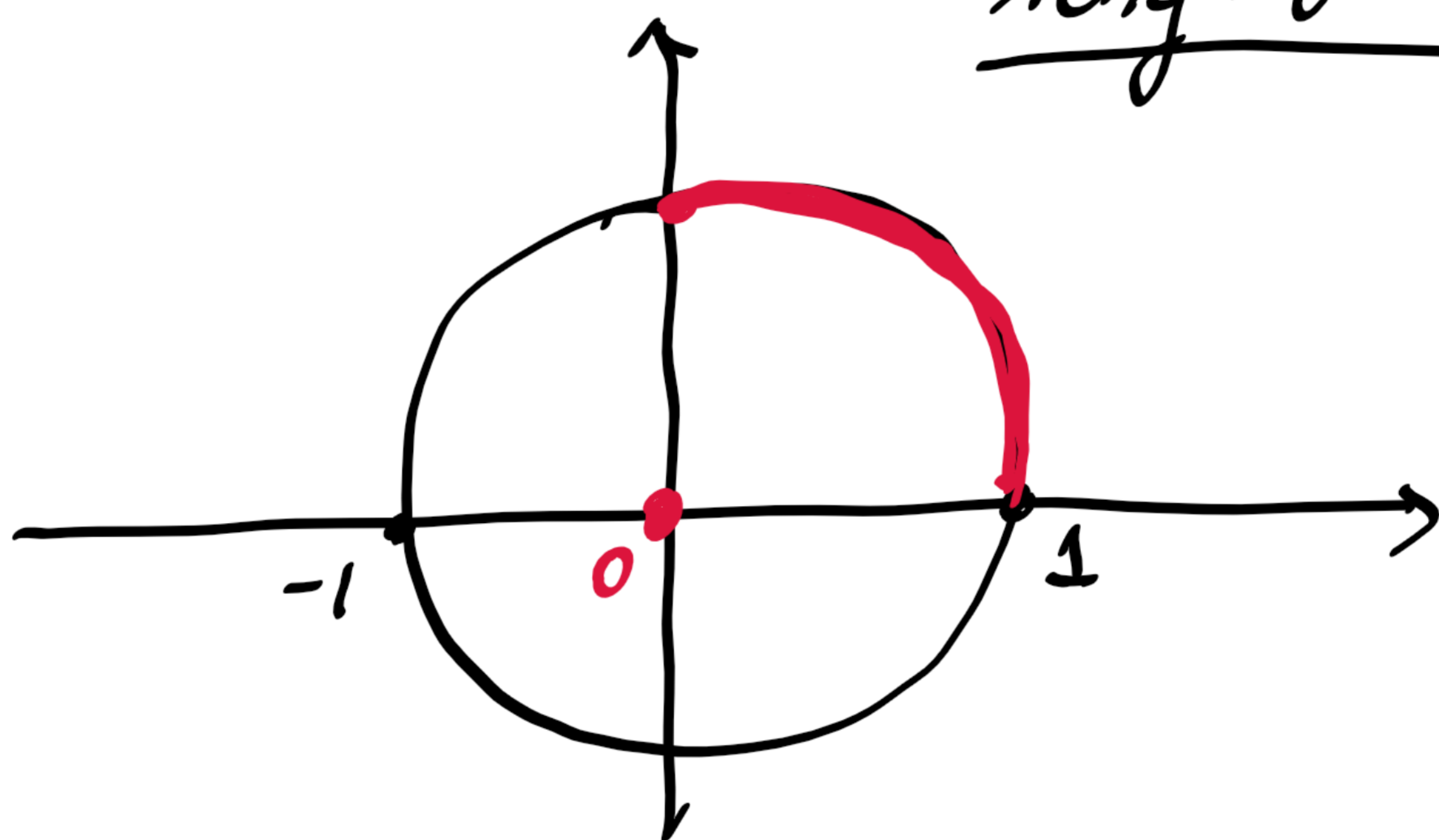
$$\pi \int_1^4 (\ln^2(x) + 4 \ln(x) + 4) dx$$

$$\int_2^{\ln(4)+2} y e^{y-2} dy$$

2.4 Arc length.



length = 2π



$$x^2 + y^2 = 1$$

$$y^2 = 1 - x^2$$

$$y = \underbrace{\sqrt{1 - x^2}}_{f(x)}$$

$$\int_0^1 \sqrt{1 + (f'(x))^2} dx$$

$$= \int_0^1 \sqrt{1 + \frac{x^2}{1 - x^2}} dx$$

$$= \int_0^1 \sqrt{\frac{\cancel{1 - x^2} + x^2}{1 - x^2}} dx \rightarrow$$

$$f'(x) = \frac{-2x}{2\sqrt{1 - x^2}} = \frac{-x}{\sqrt{1 - x^2}}$$

$$(f'(x))^2 = \frac{x^2}{1 - x^2}$$

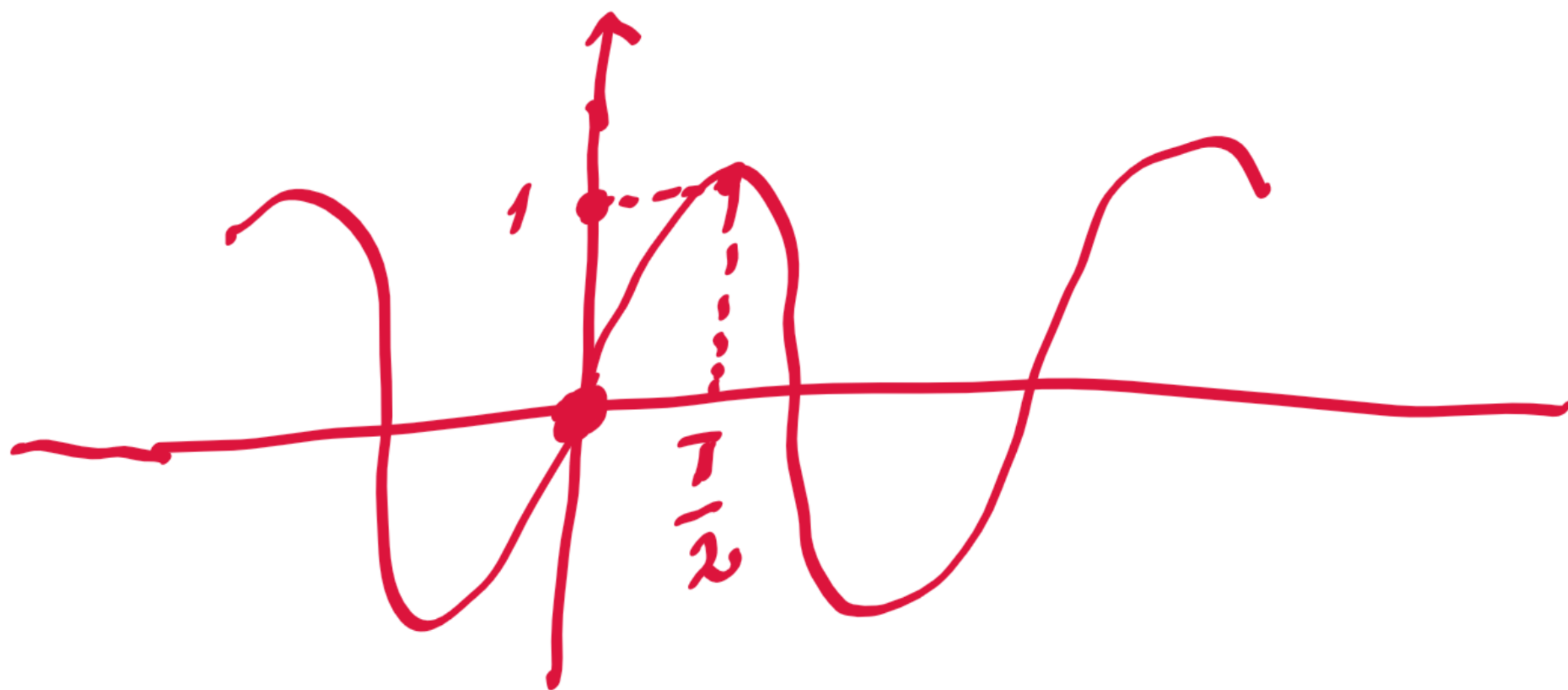
$$\int_0^1 \underbrace{\sqrt{\frac{1}{1-x^2}}}_{\frac{\sqrt{1}}{\sqrt{1-x^2}}} dx = \int_0^1 \frac{1}{\sqrt{1-x^2}} dx$$

$$= \arcsin(x) \Big|_0^1$$

$$= \arcsin(\overset{\downarrow}{1}) - \arcsin(0)$$

$$= \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

$$4 \cdot \frac{\pi}{2} = 2\pi$$



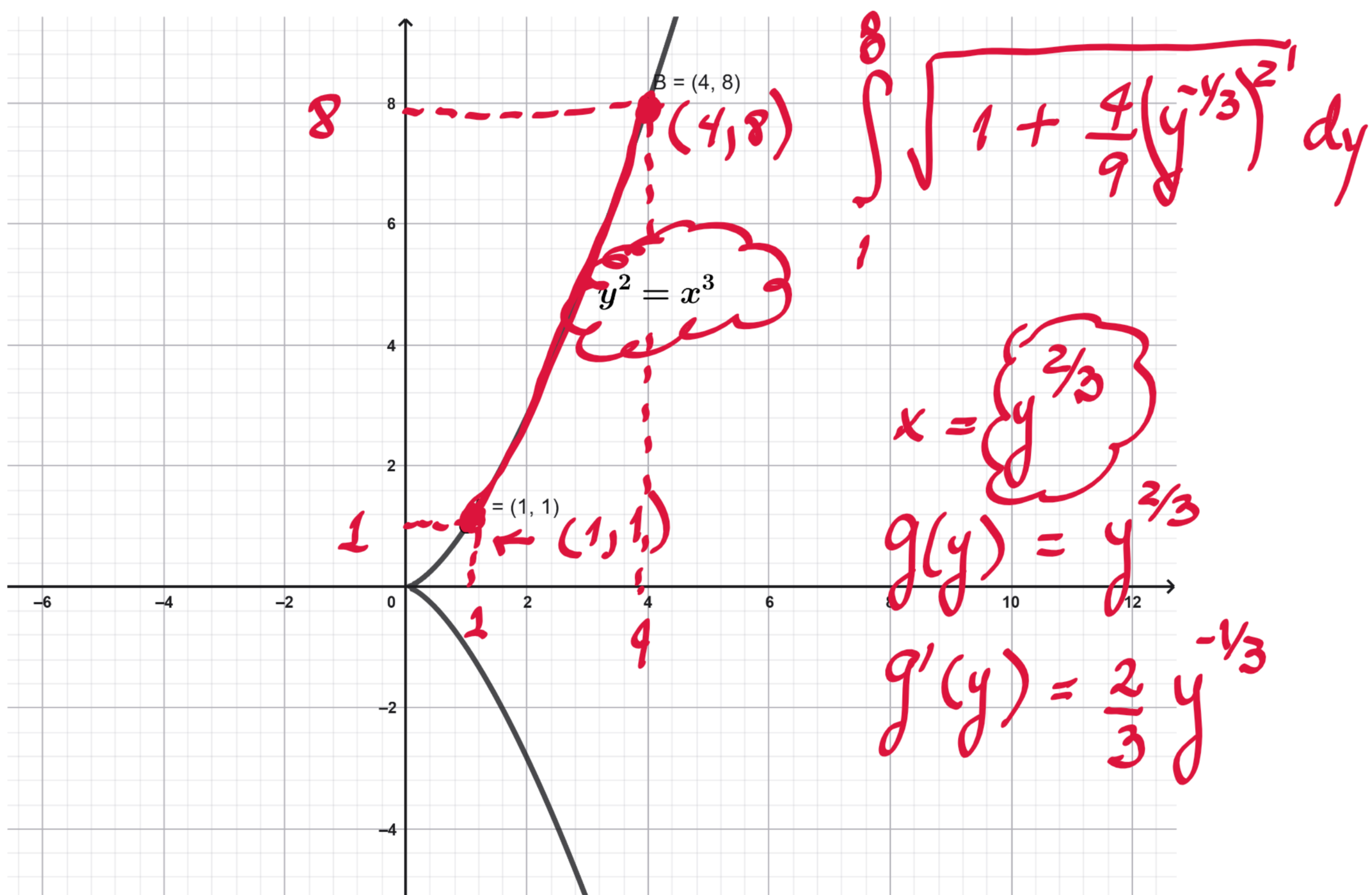
$$y^2 = x^3$$

$$\int \sqrt{1 + \frac{9}{4}x} dx$$

$$y^2 = x^3$$

$$|y| = \sqrt{x^3} = x^{3/2}$$

$$y = x^{3/2} \text{ or } y = -x^{3/2}$$



$$\int_1^8 \sqrt{1 + \frac{4}{9}(y^{1/3})^2} dy$$

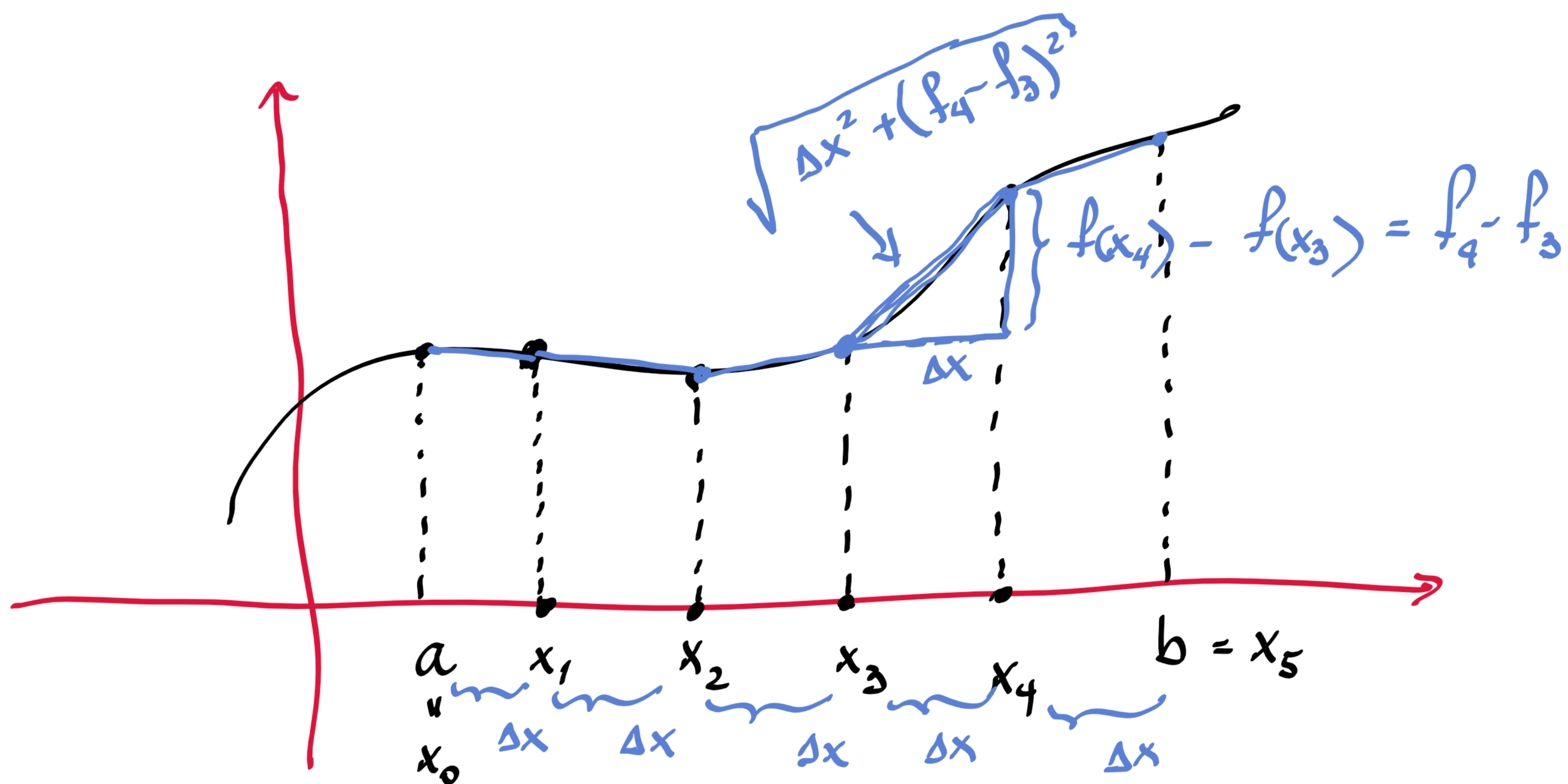
$$x = y^{2/3}$$

$$g(y) = y^{2/3}$$

$$g'(y) = \frac{2}{3} y^{-1/3}$$

$$f(x) = x^{3/2}$$

$$f'(x) = \frac{3}{2} x^{1/2}$$



$$\sum_{i=0}^{n-1} \sqrt{\Delta x^2 + (f_i - f_{i-1})^2}$$

$$\underbrace{\Delta x^2}_{\Delta x^2} + \underbrace{(f_i - f_{i-1})^2}_{\Delta x^2} = \Delta x^2 \left(1 + \left(\frac{f_i - f_{i-1}}{\Delta x} \right)^2 \right)$$

$$\sum_{i=0}^{n-1} \Delta x \sqrt{1 + \left(\frac{f_i - f_{i-1}}{\Delta x} \right)^2} \xrightarrow{n \rightarrow \infty} \int_a^b \sqrt{1 + (f'(x))^2} dx$$

Integration by parts:

$$\int (f(x) \cdot g(x))' dx = \int (f'(x) \cdot g(x) + f(x) \cdot g'(x)) dx$$

$$f(x) \cdot g(x) = \underbrace{\int f'(x) g(x) dx} + \int f(x) \cdot g'(x) dx$$

$$\underbrace{\int g(x)}_u \underbrace{f'(x) dx}_{dv} = \underbrace{f(x) \cdot g(x)}_{uv} - \underbrace{\int f(x) \cdot g'(x) dx}_{v du}$$

$$v = f(x)$$

$$dv = f'(x) dx$$

$$\int_a^b u dv = u \cdot v - \int v \cdot du$$

$$\int_a^b u dv = uv \Big|_a^b - \int_a^b v du$$

$$\int_1^4 \underbrace{\ln(x)}_u \underbrace{dx}_{dv} = x \ln(x) \Big|_1^4 - \int x \cdot \frac{1}{x} dx$$

Diff. $\swarrow$ $u = \ln(x)$
 $du = \frac{1}{x} dx$

$v = \int dv = \int dx = x$
 $dv = dx$ $\searrow$ Integ.

$$\begin{aligned}
 \int_1^4 \underbrace{\ln(x)}_u \underbrace{dx}_{dv} &= x \ln(x) \Big|_1^4 - \int_1^4 x \cdot \frac{1}{x} dx \\
 &= x \ln(x) \Big|_1^4 - \int_1^4 dx \\
 &= \underline{x \ln(x)} \Big|_1^4 - \underline{x} \Big|_1^4 = 4 \ln(4) - 1 \cdot \ln(1) - (4 - 1) \\
 &= 4 \ln(4) - 3.
 \end{aligned}$$

$$\int \underbrace{\ln^2(x)}_u \underbrace{dx}_{dv} = \int \underbrace{\ln(x)}_u \underbrace{\ln(x) dx}_{dv} \quad - \quad \underline{\text{Option 1}}$$

Option 2

$$v du = x \left(2 \ln(x) \cdot \frac{1}{x} dx \right)$$

Option 2:

$$\begin{array}{ll}
 u = \ln^2(x) & v = x \\
 du = 2 \ln(x) \cdot \frac{1}{x} dx & dv = dx
 \end{array}$$

$$\begin{aligned}
 x \ln^2(x) - \int 2 \ln(x) \cdot \frac{1}{x} \cdot \cancel{x} dx &= x \ln^2(x) - 2 \int \ln(x) dx \\
 &= x \ln^2(x) - 2 (x \ln(x) - x) + C
 \end{aligned}$$

$L | A | T^{\text{Trig}} | E \leftarrow \text{Exponential}$
 $\uparrow \quad \nwarrow \rightarrow \text{algebraic}$
 $\text{log} \quad \text{inv. trig}$

$$\textcircled{1} \int_2^{\ln(4)+2} y e^{y-2} dy$$

$$\textcircled{2} \int \arcsin(x) dx$$

$$\textcircled{3} \int x^2 e^{3x} dx$$

$$\textcircled{4} \int e^{2x} \cos(3x) dx$$

$$\textcircled{1} u = y$$

$$\downarrow$$

$$du = dy$$

$$v = e^{y-2}$$

$$dv = e^{y-2} dy$$

$$\ln(4)+2$$

$$\int_2^{\ln(4)+2} y e^{y-2} dy =$$